The next two questions pertain to the situation described below.

A ball thrown into the air lands 40 m away 2.4 s later. Assume the ball lands at the same height from which it was thrown.

1) Find the y component of the initial velocity v_{0y}

a. $v_{0y} = 41.7$ m/s
b. $v_{0y} = 11.8$ m/s
c. $v_{0y} = 23.5$ m/s
d. $v_{0y} = 16.7$ m/s
e. $v_{0y} = 47.1$ m/s

$t_{top} = 1.2s$

$V = V_{oy} - gt$

$0 = V_{oy} - g\,t_{top}$

$V_{oy} = g\,t_{top}$

$V_{oy} = 9.8(1.2)$

$V_{oy} = 11.8\ m/s$

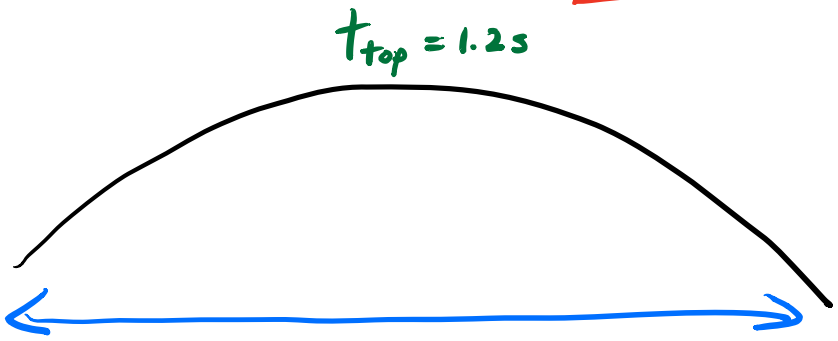
2) Find the x component of the initial velocity v_{0x}

a. $v_{0x} = 16.7$ m/s
b. $v_{0x} = 28$ m/s
c. $v_{0x} = 6.67$ m/s
d. $v_{0x} = 4.89$ m/s
e. $v_{0x} = 31.7$ m/s

$R = \Delta x = V_{ox}\,t_{total}$

$V_{ox} = \frac{R}{t_{total}} = \frac{40}{2.4}$

$V_{ox} = 16.7\ m/s$

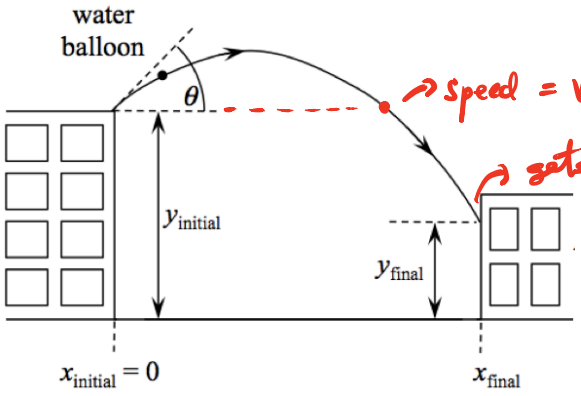


$R = \Delta x = 40m$

$t_{total} = 2.4s$

The next three questions pertain to the situation described below.

Physics students construct a cannon from rubber tubing and a funnel which will launch water-filled balloons. They assemble their device on the roof of Loomis and fire a test shot, which strikes the closest wall of the Astronomy Building, as shown in the figure.



speed = v_0

gets faster

The shot is launched from initial position $(x_{initial}, y_{initial})$ = (0 m, 16 m) and strikes the Astronomy Building at time t_{final} = 2.4 s and at the final position (x_{final}, y_{final}) = (180 m, 8 m).

3) The water balloon's speed is greatest

a. When it reaches its maximum altitude.
b. When it is launched.
c. When it strikes the Astronomy Building.

4) Which of the following best describes the magnitude of the vertical component of the water balloon's velocity during its flight towards the Astronomy Building?

a. It decreases then it increases.
b. It is constant.
c. It increases then it decreases.

5) What is the initial vertical component of the balloon's velocity, v_{0y}?

a. v_{0y} = 75 m/s
b. v_{0y} = 3.33 m/s
c. v_{0y} = 0 m/s
d. v_{0y} = 8.44 m/s
e. v_{0y} = 23.54 m/s

$\Delta y = -8$ m

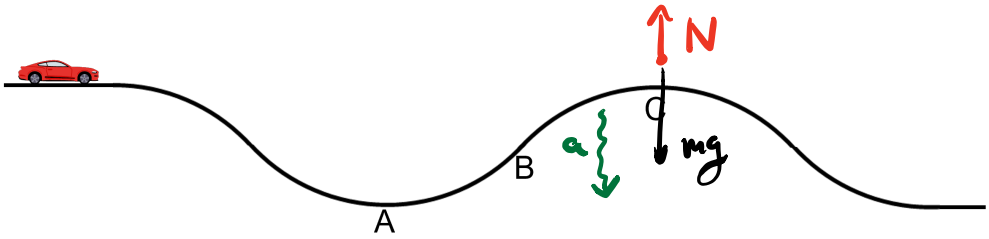
$\Delta y = v_{0y} t - \frac{1}{2} g t^2$

$v_{0y} = \dfrac{\Delta y + \frac{1}{2} g t^2}{t}$

$v_{0y} = \dfrac{-8 + \frac{1}{2}(9.8)(2.4)^2}{2.4}$

$v_{0y} = 8.4$ m/s

The next two questions pertain to the situation described below.



Your are driving along a road when you arrive at the top of an interesting hill. The bottom of the hill appears to form a semicircle with radius $R = 50$ m, and it leads up another hill whose top is also semicircular with the same radius. Your car is set to maintain a constand speed of 20 m/s. The total mass of the car including occupants is 1100 kg.

6) What is the magnitude of the acceleration of the car when it is at the bottom of the hill (point **A**)

a. $a_A = 8$ m/s^2
b. $a_A = 0$ m/s^2
c. $a_A = 18$ m/s^2

$$a = \frac{v^2}{R} = \frac{(20)^2}{50} \Rightarrow a = 8\ m/s^2$$

7) What is the force of the road on the car when it is at the top of the hill (point **C**) ?

a. F = 8800 N
b. F = 1.1×10^4 N
c. F = 2×10^3 N

$$mg - N = \frac{mv^2}{R}$$

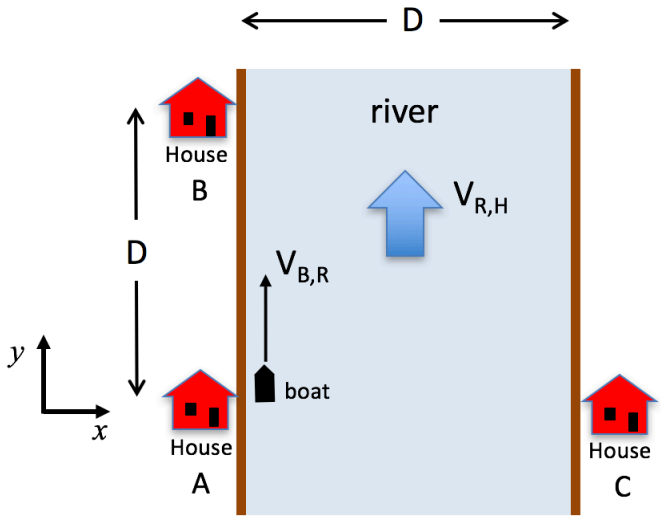
$$N = m\left(g - \frac{v^2}{R}\right)$$

$$N = 1100(9.8 - 8)$$

$$N = 1980 \approx 2000N$$

The next three questions pertain to the situation described below.

The houses, A, B , and C, are located on a river as shown. House B is on the same side of the river as House A, and is a distance $D = 400$ m away. House C is directly across the river from House A, and is also a distance $D = 400$ m away. The river flows in the +y direction with speed $V_{R,H} = 2.6$ m/s relative to the houses. Your boat can move with maximum velocity of $V_{B,R} = 4.3$ m/s relative to the water of the river.



8) What is Δt_{AB}, the shortest time in which you can drive your boat from House A to House B ?

a. $\Delta t_{AB} = 93$ s
b. $\Delta t_{AB} = 58$ s
c. $\Delta t_{AB} = 153.8$ s
d. $\Delta t_{AB} = 235.3$ s
e. $\Delta t_{AB} = 116.8$ s

$V_{B,H} = V_{B,R} + V_{R,H}$

$= 4.3 + 2.6 = 6.9$ m/s

$\Delta t = \frac{D}{V_{B,H}} = \frac{400}{6.9} \Rightarrow \boxed{\Delta t = 58\text{s}}$

9) If you start at House A and direct your boat directly across the river, how far will you be from House C when you get to the other side of the river?

a. $y_C = 242$ m
b. $y_C = 304$ m
c. $y_C = 151$ m

in x

$\Delta x = V_{B,R} t$

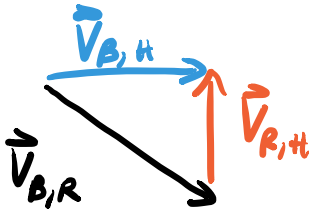
$t = \frac{400}{4.3} =$

in y

$\Delta y = V_{R,H} t = (2.6)\left(\frac{400}{4.3}\right) \Rightarrow \boxed{242\text{ m}}$

10) What is Δt_{AC}, the shortest time in which you can drive your boat from House A to House C ?

a. $\Delta t_{AC} = 116.8$ s
b. $\Delta t_{AC} = 235.3$ s
c. $\Delta t_{AC} = 58$ s



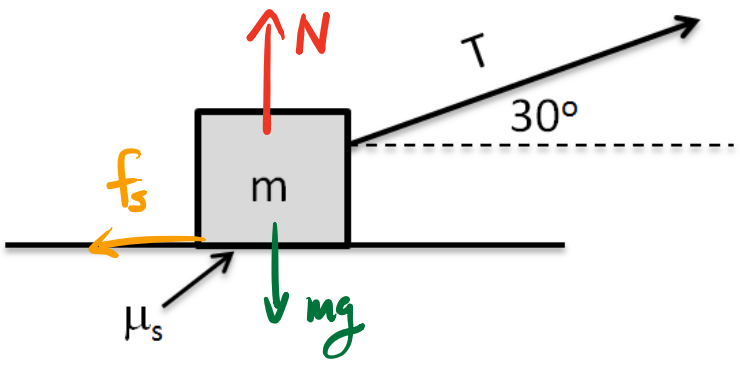
$V_{B,H} = \sqrt{(4.3)^2 - (2.6)^2}$

$= 3.42$ m/s

$\Delta t = \frac{D}{V_{B,H}} = \frac{400}{3.42}$

$\boxed{\Delta t = 116.9\text{s}}$

The next three questions pertain to the situation described below.



A block of mass m = 1.2 kg, at rest on a level surface, is being pulled by a string having tension T = 3 N at an angle of 30° above the horizontal, as shown in the diagram. Static friction keeps the block from moving. The coefficient of static friction between the block and the surface is $\mu_S = 0.45$.

11) The normal force acting on the block is

a. Less than the weight of the block.
b. Greater than the weight of the block.
c. Equal to the weight of the block.

12) What is the magnitude of the static frictional force on the block?

a. 5.3 N
b. 1.5 N
c. 3 N
d. 4.6 N
e. 2.6 N

$$\Sigma F_x = 0 = T\cos\theta - f_s = 0$$

$$f_s = T\cos\theta = 3\cos 30°$$

$$f_s = 2.6\,N$$

13) Now assume that there is no friction, with everything else in the problem remaining the same. What is the acceleration of the block?

a. 2.5 m/s^2
b. 2.2 m/s^2
c. 1.3 m/s^2
d. 0 m/s^2
e. 7.3 m/s^2

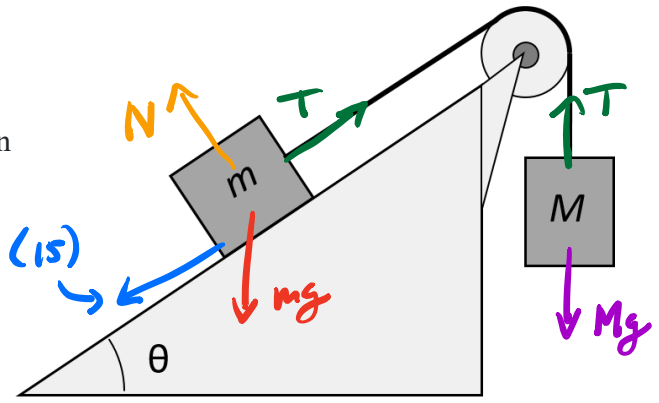
$$T\cos\theta = ma$$

$$a = \frac{T\cos\theta}{m} = \frac{3\cos 30°}{1.2}$$

$$a = 2.2\ m/s^2$$

The next three questions pertain to the situation described below.

A block of mass m is at rest on a frictionless inclined plane at an angle $\theta = 30°$ from the horizontal, as shown in the figure. The block m is connected to a second block of mass M by a massless string around an ideal, frictionless pulley.



14) What must the mass M be for the blocks to remain at rest?

a. $M = m$
b. $M = 2m$
c. $M = m/2$

$Mg = T = mg\sin\theta$

$M = m\sin\theta = m\sin 30 \Rightarrow \boxed{M = \frac{m}{2}}$

15) Now suppose that we add friction to the surface of the inclined plane, using material with coefficient of static friction $\mu_s = 0.4$ and kinetic friction $\mu_k = 0.3$. What is the maximum the mass M can be for the blocks to remain at rest?

a. $M_{max} = 0.15m$
b. $M_{max} = 0.5m$
c. $M_{max} = 0.65m$
d. $M_{max} = 0.85m$
e. $M_{max} = 1.65m$

add $f_{s,max}$ (down ramp)

$Mg = mg\sin\theta + f_{s,max} = mg\sin\theta + \mu_s mg\cos\theta$

$M = \frac{m}{2} + 0.4\cos 30\, m \Rightarrow 0.85m$

16) If the block on the inclined plane had not been at rest but moving at a speed v initially, and we want it to maintain a constant speed, the mass M should be _____ compared to the previous problem.

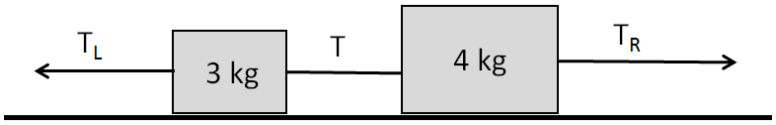
a. the same
b. increased
c. decreased

if moving use kinetic friction $f_k < f_{s,max}$

We want to ↓ tension

Decrease weight of hanging Mass

The next three questions pertain to the situation described below.



Two blocks on a level frictionless surface have mass 3 kg and 4 kg. A string pulls the 3 kg block to the left with tension $T_L = 4.3$ N, a second string pulls the 4 kg block to the right with tension $T_R = 14.3$ N. The two blocks are connected together by a third string in the middle as shown.

17) What is the tension in the middle string, labeled T in the diagram?

a. T = 4.3 N
b. T = 8.6 N
c. T = 14.3 N
d. T = 18.6 N
e. T = 10 N

1) $T - T_L = 3a$
2) $-T + T_R = 4a$
} solve for T

$T_R - T_L = 7a$

$a = \frac{T_R - T_L}{7} = 1.43 \text{ m/s}^2$

$T = 3a + T_L$

$T = 3(1.43) + 4.3$

$T = 8.6\text{N}$

18) Suppose the mass of both blocks was doubled but T_L and T_R were kept the same. How would the tension in the middle string change?

a. T would increase.
b. T would decrease.
c. T would stay the same.

see equation above

$T = 3a + T_L$

$3 \to 6$

but $a \to \frac{a}{2}$

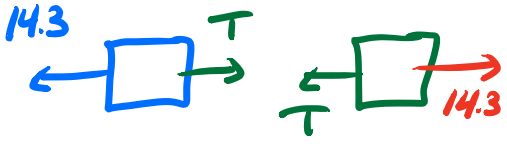
$T = \frac{6a}{2} + T_L$

No change

19) Suppose T_L and T_R were both set to 14.3 N. The tension in the middle string would be

a. T = 0 N
b. T = 14.3 N
c. T = 28.6 N

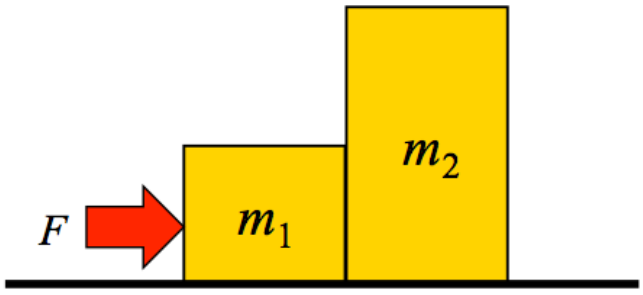
acceleration of each block = 0



T must be 14.3N

The next three questions pertain to the situation described below.

A box of mass $m_1 = 2$ kg is in contact with a heavier box of mass $m_2 = 4$ kg. The boxes are initially at rest on a horizontal surface, and the coefficients of kinetic and static friction between the boxes and the surface are $\mu_K = 0.15$ and $\mu_S = 0.25$, respectively. A horizontal force F is applied to the left of m_1 as shown in the figure.



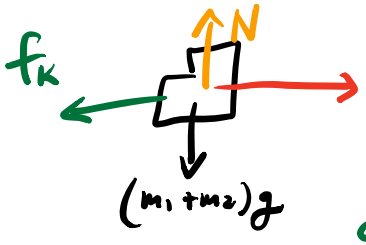
20) How large must F be to set the boxes in motion?

a. 5.89 N
b. 8.83 N
c. 0 N
d. 2.94 N
e. 14.72 N

$f_{s,max} = \mu_s N_{TOTAL}$
$= \mu_s (m_1 + m_2) g$
$= 0.25(6)(9.8)$
$f_{s,max} = 14.7\,N$

21) When $F = 22.07$ N, what is the acceleration of the boxes?

a. 11.04 m/s^2
b. 2.21 m/s^2
c. 3.68 m/s^2
d. 1.23 m/s^2
e. 5.52 m/s^2



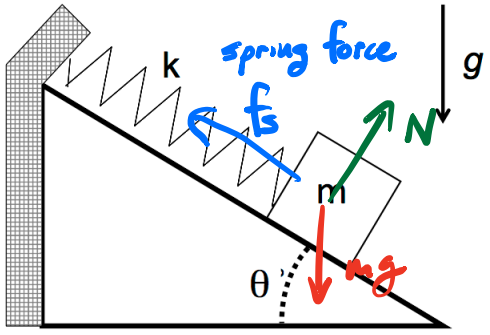
$F - f_K = m_{TOT} a$
$a = \frac{F - \mu N}{m_{TOT}} = \frac{22.07 - 0.15(6)(9.8)}{6}$
$a = 2.2\,m/s^2$

22) When $F = 22.07$ N, what is F_{net2}/F_{net1}, the ratio of the net forces acting on m_2 and m_1?

a. $F_{net2}/F_{net1} = m_1/m_2$
b. $F_{net2}/F_{net1} = m_2/(m_1 + m_2)$
c. $F_{net2}/F_{net1} = m_2/m_1$

$F_{NET2} = m_2 a$
$F_{NET1} = m_1 a$
$\frac{F_{NET,2}}{F_{NET,1}} = \frac{m_2}{m_1}$

An ideal spring holds a stationary mass on a frictionless plane inclined by θ = 26 degrees to the horizontal. The spring constant is k = 200 N/m and the mass m is 2 kg.



23) By what distance, Δx, is the spring stretched from its equilibrium length?

a. 1.5 cm
b. 4.3 cm
c. 8.8 cm
d. 6.2 cm
e. 13.4 cm

$$\Sigma F_x = 0$$

$$mg\sin\theta - f_s = 0$$

$$mg\sin\theta - k\Delta x = 0$$

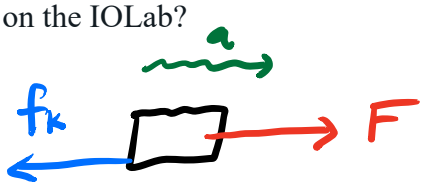
$$\Delta x = \frac{mg\sin\theta}{k} = \frac{2(9.8)\sin 26}{200}$$

$$\Delta x = 0.043\,m = \boxed{4.3\,cm}$$

In a lab experiment, you attach a string to the force sensor of an IOLab. The mass of the IOLab is 0.19 kg. You set the IOLab on its wheels on the surface of a horizontal table and pull the IOLab straight across the table with the string. Data from the IOLab's force sensor show that you were pulling with a constant force of 0.9 N. Data from the IOLab's wheel sensor indicates it started from rest and had a velocity of 1.3 m/s after it has travelled 39 cm.

24) What was the force of friction acting on the IOLab?

a. 0.58 N
b. 0.9 N
c. 0.49 N



$$F - f_k = ma$$

$$f_k = F - ma$$

Need to find a

use kinematics

$$V^2 = V_0^2 + 2a\Delta x$$

$$a = \frac{V^2}{2\Delta x}$$

$$f_k = F - m\left(\frac{V^2}{2\Delta x}\right)$$

$$f_k = 0.9 - 0.19\left[\frac{1.3^2}{2(0.39)}\right]$$

$$f_k = 0.49N$$

This is the last question of the exam!

25) Did you check your answer sheet to ensure you

1. Entered your netid
2. Entered your exam version
3. Marked answers for all 25 questions (including this one)

a. Yes (I did all 3)
b. No (I did not enter my netid)
c. No (I did not enter my exam version)